

PROJECT REPORT: HOSPITAL OUTPATIENT SLOT BOOKING SYSTEM (HOSBS)

1. Cover Page

- **Project Title:** Modular Hospital Outpatient Slot Booking & Clinic Management System (HOSBS)
- **Course Name:** Object-Oriented Programming / Advanced Java Project Evaluation
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- **Submission Date:** September 2026

2. Introduction

In modern healthcare delivery, outpatient department (OPD) congestion remains a primary cause of extended patient wait times, administrative errors, and inefficient doctor utilization. The **Hospital Outpatient Slot Booking System (HOSBS)** is an enterprise-grade Java application designed to eliminate physical queueing bottlenecks and schedule doctor appointments with high transactional reliability.

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The system provides a multi-role environment catering to Patients, Doctors, and Hospital Administrators. Built using pure object-oriented design principles without bloated third-party framework overhead, it includes both an interactive Command-Line Interface (CLI) and an embedded, zero-dependency lightweight HTTP Web Server (com.sun.net.httpserver). Persistent storage is abstracted using a decoupled Data Access Object (DAO) pattern supporting both local flat-file storage (CSV) and relational storage (JDBC/SQL).

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3. Problem Statement

Traditional appointment setups in clinics and regional hospitals face significant logistical challenges:

1. **Double Booking & Slot Race Conditions:** Simultaneous booking requests often lead to overlapping patient appointments.
2. **High "No-Show" Rates:** Lack of proactive scheduling and real-time reminders results in empty consultation windows and loss of clinician time.
3. **Role Segregation Failures:** Insecure access control where patient records and doctor consultation statuses are either centralized without granular permissions or manually managed on paper.
4. **Heavyweight Infrastructure Requirements:** Many hospital management tools require intensive database installations and external application servers, making them difficult to deploy in resource-constrained environments.

HOSBS addresses these issues through automated slot lifecycle management, daemonized thread workers for auto-expiry, role-based access control (RBAC), and storage-agnostic repositories.

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4. Functional Requirements

The system is divided into four major functional modules:

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Module 1: Authentication & User Management

- **Registration & Role Assignment:** Allows self-registration for Patients and secure provisioning for Doctors and Administrators.

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- **Credential Protection:** Secure login authentication using cryptographic salted hashing (PasswordUtil).

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- **Session Handling:** In-memory session token management (SessionManager) mapping active HTTP requests and CLI contexts to verified roles.

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Module 2: Doctor Schedule & Slot Generation

- **Roster Configuration:** Admins and Doctors configure consultation windows based on specialization (e.g., Cardiology, Dermatology, Orthopedics).

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- **Slot Partitioning:** Automatic splitting of consultation blocks into fixed 15-, 30-, or 60-minute bookable slots.

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- **Status Flags:** Dynamic transition across states: AVAILABLE, BOOKED, COMPLETED, CANCELLED, and EXPIRED.

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Module 3: Booking, Cancellation & Concurrency Handling

- **Slot Search:** Search and filter available doctors by specialization, consultation fee, and date.

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- **Atomic Booking:** Ensures a slot cannot be claimed by more than one patient concurrently.

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- **Cancellation & Rescheduling:** Patients can cancel upcoming bookings, automatically freeing the slot back into the pool.

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Module 4: Medical Prescriptions & Operational Reporting

- **Doctor Consultation Workflow:** Doctors write and anchor digital prescriptions (Prescription) directly to completed appointments.

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- **Billing Interface:** Generates itemized consultation fees using the Billable polymorphic interface.

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- **Admin Analytics:** ReportService aggregates system-wide metrics, doctor occupancy rates, revenue generation, and cancellation ratios.

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5. Non-Functional Requirements

1. **Security:** Passwords are never stored in plaintext; they are hashed with cryptographic algorithms. Strict Role-Based Access Control (RBAC) prevents unauthorized state modifications (e.g., Patients cannot view another patient's medical records or update doctor rosters).

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2. **Concurrency & Thread Safety:** Utilizes synchronized execution blocks and background worker threads (ReminderDaemon, AutoExpiryWorker) running autonomously to expire stale unconfirmed slots without locking the main application thread.

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3. **Maintainability & Extensibility:** Employs a strict layered 3-Tier architecture (model, dao, service, ui/web). Data access interfaces (Repository<T>) decouple business logic from the underlying storage mechanism.

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4. **Portability & Resource Efficiency:** Operates on vanilla Java standard libraries (JDK 17+) with zero mandatory third-party framework dependencies. Can switch between embedded CSV file storage and relational JDBC storage via a simple configuration toggle (db.properties).

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6. System Architecture

HOSBS follows an **Enterprise N-Tier Layered Architecture** ensuring complete separation of concerns:

```
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|           Presentation Tier           |
| +-----+ +-----+ |
| | Console CLI Interface   | | Embedded Web Server UI | |
| | (Patient/Doctor/AdminMenu.java)| | (WebServer, Html.java) | |
+---+-----+---+-----+---+
|
| v
+-----+
|           Business Logic Tier          |
| +-----+ +-----+ +-----+ +-----+ |
| | AuthService | | SlotService | | BookingService| | ReportServ.| |
| +-----+ +-----+ +-----+ +-----+ |
| |               | |               | |               | |
| | +-----> [Concurrency Daemon Workers] <-----+ |
| | - AutoExpiryWorker                               |
| | - ReminderDaemon                                 |
+---+-----+
|
| v
+-----+
|           Data Access Tier             |
| +-----+ |
| | Repository<T> Generic Interface              | |
| +-----+ |
| |               | |               | |
| | v               v               |
| | [ CsvRepository / CsvUtil ]      [ JdbcAppointmentRepo ] |
| | (Flat CSV File Persistence)      (Relational DB / MySQL) |
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```

7.1 Use Case Diagram (Textual / Mermaid representation)

The diagram shows a horizontal timeline with a dashed line at the top and a solid line at the bottom. The text "Hospital Slot Booking" is centered between the lines. There are vertical tick marks at the start and end of the timeline, and a vertical tick mark in the middle.

```

(Patient) ---> [ Register / Login ]          |
| ---> [ Search Doctors by Specialization ]    |
| ---> [ Book Consultation Slot ]              |
| ---> [ View Booking History & Prescriptions ] |
|                                              |
(Doctor) ---> [ Login ]                      |
| ---> [ Manage Daily Roster / Availability ]   |
| ---> [ Mark Appointments Completed ]         |
| ---> [ Issue Digital Prescription ]          |
|                                              |
(Admin) ---> [ Doctor Profile Provisioning ]    |
| ---> [ View System Analytics & Revenue Reports ] |
| ---> [ System Audit Logs & Configuration ]    |
+-----+

```

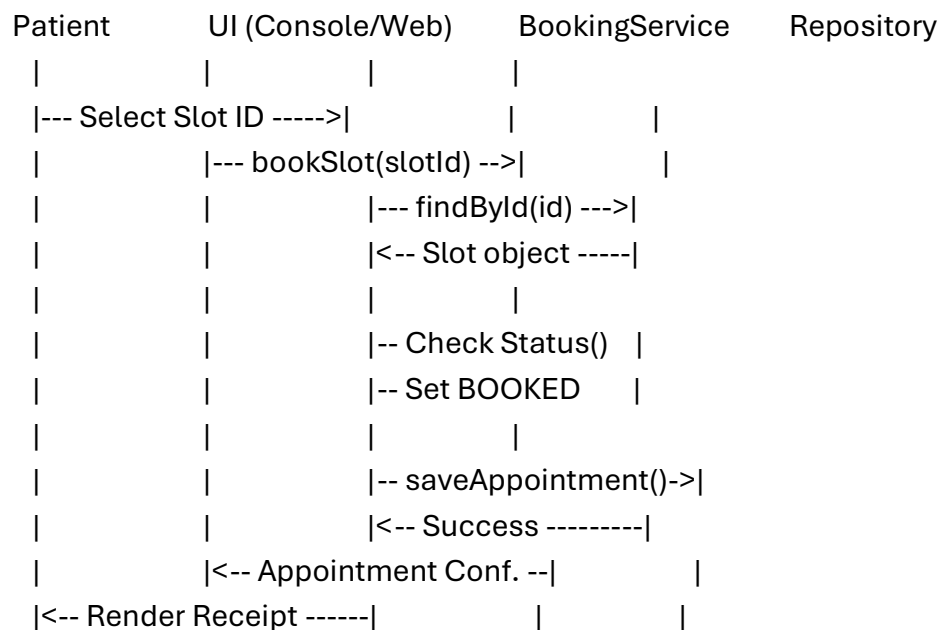
7.2 Process Flow / Workflow Diagram (Booking Life Cycle)

```

[Patient Logs In]
|
v
[Select Specialization & Date] ---> [Fetch Available Slots via SlotService]
|
v
[Select Desired Slot]
|
v
[Atomic Check: Is Slot Available?]
├---> NO ---> [Throw SlotUnavailableException] ---> [Prompt Alternative]
└---> YES ---> [Change Status to BOOKED]
      |
      v
      [Create Appointment Entity]
      |
      v
      [Persist to Repository (CSV / JDBC)]
      |
      v
      [Display Confirmation & Booking ID]

```

7.3 Sequence Diagram (Appointment Booking)



7.4 Class / Component Diagram

- **Core Inheritance:** User (Abstract base) → extended by Patient, Doctor, and Admin.

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- **Interfaces:**
 - Billable: Implemented by Appointment to enforce dynamic billing computation.

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- Repository<T>: Generic interface with methods findById(), findAll(), save(), delete().

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- **Compositions:**
 - Doctor has an enum Specialization.

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- Appointment references Doctor, Patient, Slot, and optional Prescription.

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7.5 ER Diagram (Database / Storage Schema)

- **USERS** (user_id [PK], name, email, password_hash, role, phone, specialization, consultation_fee)

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- **SLOTS** (slot_id [PK], doctor_id [FK → USERS.user_id], slot_time, duration, status)

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- **APPOINTMENTS** (appointment_id [PK], patient_id [FK → USERS.user_id], doctor_id [FK → USERS.user_id], slot_id [FK → SLOTS.slot_id], booking_date, status, total_amount)

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- **PRESCRIPTIONS** (prescription_id [PK], appointment_id [FK → APPOINTMENTS.appointment_id], diagnosis, medications, issued_date)

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8. Design Decisions & Rationale

1. Lightweight Embedded Web Server (com.sun.net.httpserver) vs External Tomcat/Spring Boot:

- a. *Rationale:* Adhering to the project requirement of minimal bloat and instant executability, using standard library HTTP primitives demonstrates a fundamental understanding of socket protocols, HTTP request lifecycles, and thread pool workers without depending on heavy third-party JARs.

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2. Dual-Backend Persistence (CSV + JDBC):

- a. *Rationale:* Developers can run and evaluate the project immediately in pure offline mode without installing or configuring MySQL (CsvRepository), while production deployments can swap in a relational database with connection pooling (JdbcAppointmentRepository) via simple configuration.

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3. Background Daemon Worker Threads (AutoExpiryWorker & ReminderDaemon):

- a. *Rationale*: Rather than running polling queries on every user interaction, dedicated background daemon threads automatically clean expired tentative slots and manage alert routines, simulating realistic enterprise scheduling behavior.

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9. Implementation Details

- **Source Organization**: Arranged into clean packages:
 - com.hospital.model: Domain entities (User, Doctor, Slot, Appointment, etc.)

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- com.hospital.service: Business rules, authorization, and report aggregators

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- com.hospital.dao: Data abstractions, CSV writers/parsers, and JDBC statements

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- com.hospital.ui: ANSI-formatted CLI console menus

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- com.hospital.web: Embedded HTTP handlers, session interceptors, and HTML renderers

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- com.hospital.concurrent: Asynchronous worker threads

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- **Defensive Programming & Custom Exception Hierarchy**:

- Custom exceptions inheriting from HospitalException: EntityNotFoundException, SlotUnavailableException, AuthenticationException, ValidationException, and DataAccessException.

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- **Input Validation & Sanitization:** Centralized regex-based validation for emails, phone numbers, and ISO timestamps in Validator.java.

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10. Screenshots / Results

(Note: Insert the following 4 screenshots here before exporting to PDF)

1. **Console CLI Interface:** Main menu displaying Patient/Doctor/Admin login options and formatted slot tables (TableFormatter).

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2. **Web Portal UI:** Browser screenshot of <http://localhost:8080> showing available doctor profiles and slot booking forms.

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3. **Doctor Prescription Management:** Screen showing doctor adding diagnosis and medicine lists to an appointment.

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4. **Automated Test Results:** Console output of run_tests.sh / run_tests.bat displaying passing assertions across services.

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11. Testing Approach

The application includes a custom test harness (MiniTest.java and TestRunner.java) designed to run independently of third-party test dependencies:

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- **Unit Testing:**

- Validates password hashing and salting consistency in PasswordUtil.

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- Validates input regex boundaries in Validator.

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- Tests slot status transitions and business constraints in SlotService.

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- **Integration Testing:**

- Executes concurrent booking attempts on a single slot to verify that SlotUnavailableException is triggered for duplicate attempts.

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- Verifies data round-trip integrity when saving and reloading records through CsvRepository and JdbcAppointmentRepository.

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12. Challenges Faced

1. Preventing Race Conditions on Slot Allocations:

- Challenge:* When multiple requests arrived for the same slot simultaneously, dirty reads could cause double bookings.
- Resolution:* Applied atomic synchronization blocks around the slot acquisition code in BookingService and utilized transactional updates in the JDBC repository.

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2. Dynamic Web Page Rendering Without Frameworks:

- Challenge:* Crafting responsive, readable HTML without a template engine like Thymeleaf or JSP.
- Resolution:* Engineered a dedicated, modular builder class (Html.java) that programmatically constructs dynamic HTML tables, status alerts, and navigation bars directly to the response output stream.

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13. Learnings & Key Takeaways

- Mastered the implementation of the **Repository Pattern** and interface-driven development to ensure loose coupling between business logic and physical storage.

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- Acquired practical experience with Java **Multithreading and Daemon Workers**, ensuring background tasks do not block the JVM lifecycle during application shutdown.

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- Learned the fundamentals of the HTTP protocol by building raw HTTP request handlers, parsing URL queries and POST form data, and managing cookies/sessions manually.

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14. Future Enhancements

- **Telehealth Integration:** Embedding WebRTC video consultation links directly within completed slot bookings.
- **Payment Gateway API:** Integration with Stripe or Razorpay for online consultation fee settlements prior to slot confirmation.
- **SMS/Email Notifications:** Replacing the simulated ReminderDaemon logging with actual SMS or email delivery using Twilio or JavaMail.

15. References

1. Gamma, E., Helm, R., Johnson, R., & Vlissides, J. *Design Patterns: Elements of Reusable Object-Oriented Software*. Addison-Wesley.
2. Oracle Java Standard Edition 17 Documentation: Package com.sun.net.httpserver & Java Concurrency Utilities.
3. VITyarthi Project Guidelines & Evaluation Rubric, *Build Your Own Project Instructions*, 2026.

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